COURSE CODE : ITA0608

COURSE NAME : MACHINE LEARNING

Experiment 12: Implement Iris Flower Classification using K-Nearest Neighbors (KNN)

Aim

To implement the K-Nearest Neighbors (KNN) algorithm for classifying Iris flowers into Setosa, Versicolor, and Virginica species based on their sepal and petal measurements.

Algorithm

1. Import the required libraries.
2. Load the Iris dataset.
3. Separate the dataset into features (X) and target labels (y).
4. Split the dataset into training and testing sets.
5. Create a K-Nearest Neighbors (KNN) classifier with an appropriate value of **k** (e.g., k = 3).
6. Train the classifier using the training dataset.
7. Predict the species for the testing dataset.
8. Evaluate the model using accuracy score and classification report.
9. Predict the class of a new flower sample.

Code

```python
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, classification_report
iris = load_iris()
X = iris.data
y = iris.target
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
knn = KNeighborsClassifier(n_neighbors=3)
knn.fit(X_train, y_train)
y_pred = knn.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
print("\nClassification Report:")
print(classification_report(y_test, y_pred, target_names=iris.target_names))
new_sample = [[5.1, 3.5, 1.4, 0.2]]
prediction = knn.predict(new_sample)
print("\nPredicted Flower Species:", iris.target_names[prediction][0])
```

Output

```
Accuracy: 1.0

Classification Report:
              precision    recall  f1-score   support

      setosa       1.00      1.00      1.00        10
  versicolor       1.00      1.00      1.00         9
   virginica       1.00      1.00      1.00        11

    accuracy                           1.00        30
   macro avg       1.00      1.00      1.00        30
weighted avg       1.00      1.00      1.00        30


Predicted Flower Species: setosa
```

Result

The K-Nearest Neighbors (KNN) algorithm was successfully implemented for Iris flower classification. The model classified the flowers into Setosa, Versicolor, and Virginica with high accuracy (approximately 97%–100% on the Iris dataset), demonstrating that KNN is an effective supervised machine learning algorithm for multi-class classification tasks.